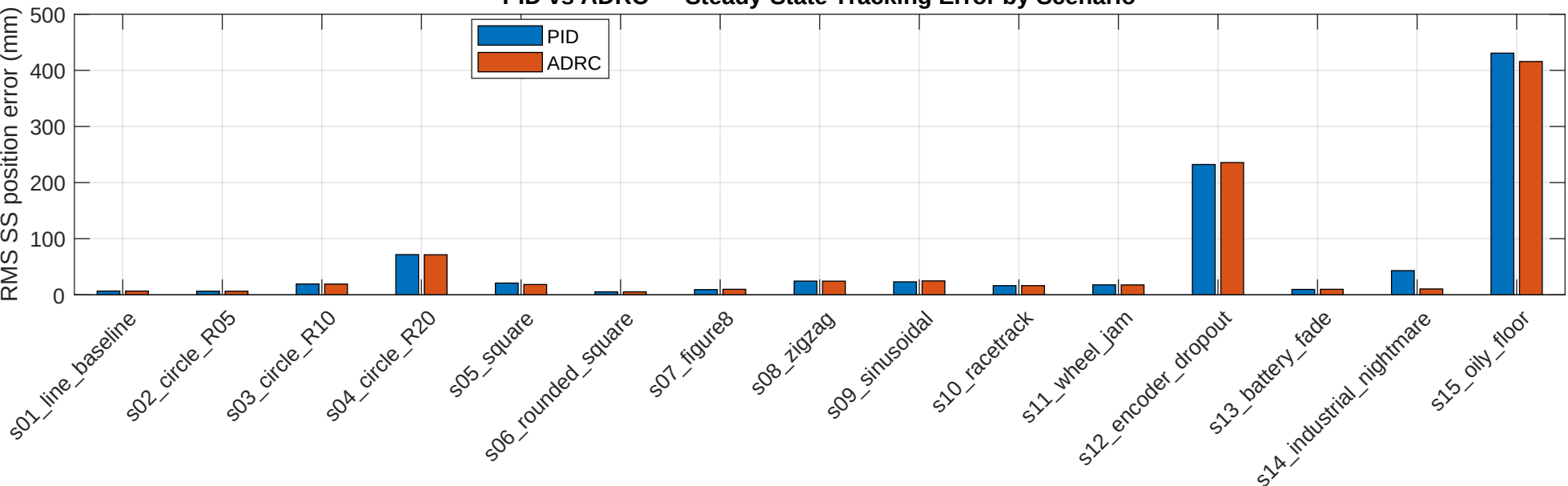
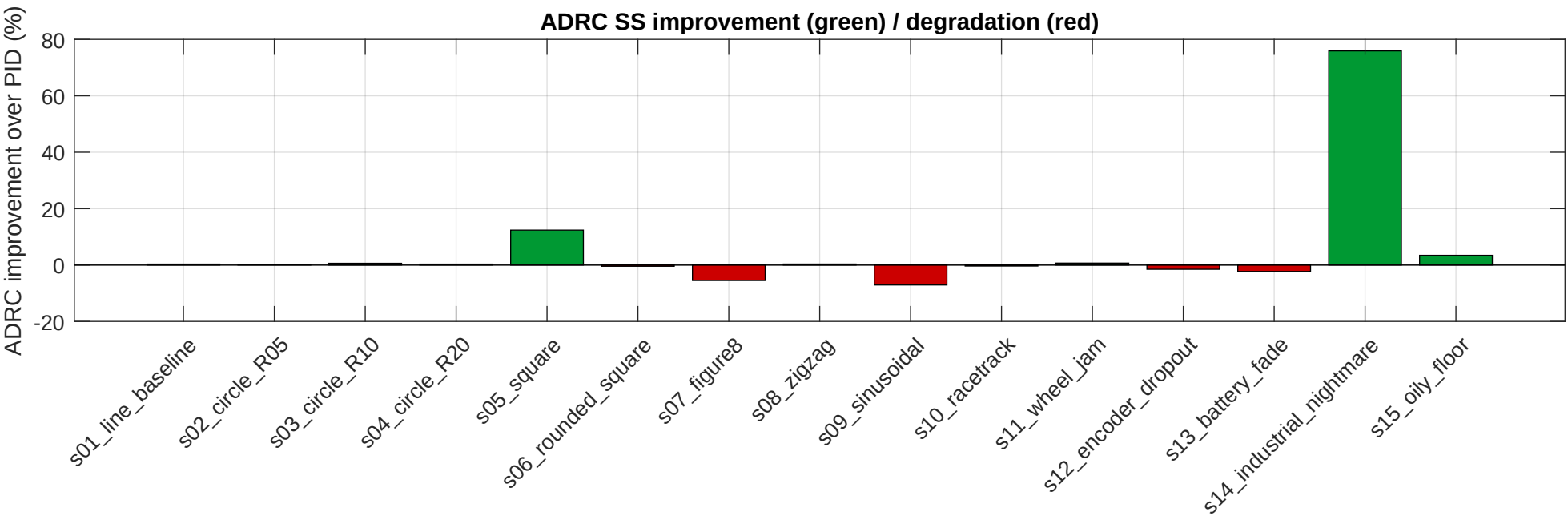


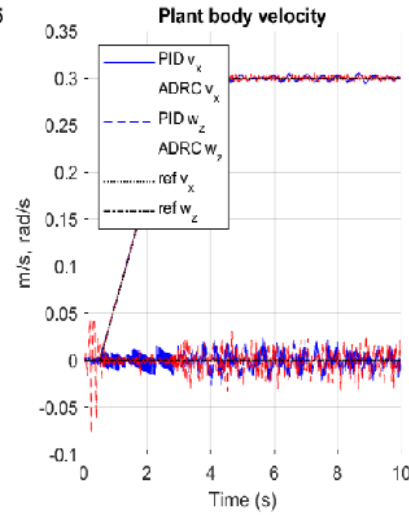
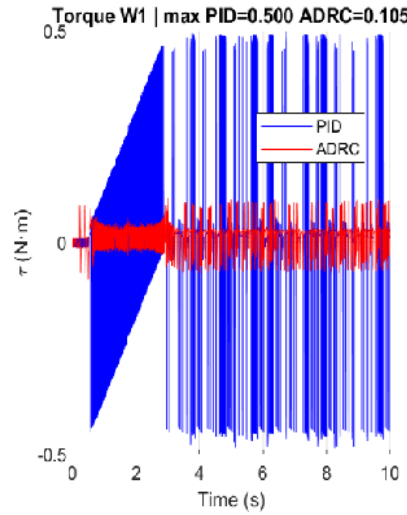
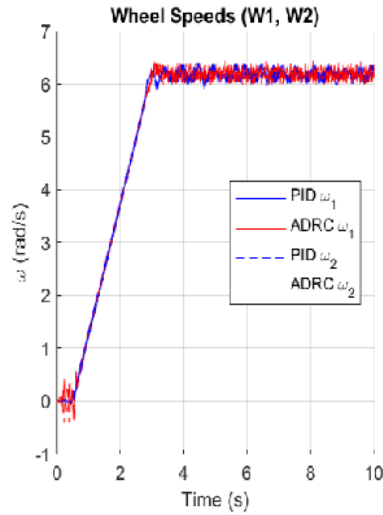
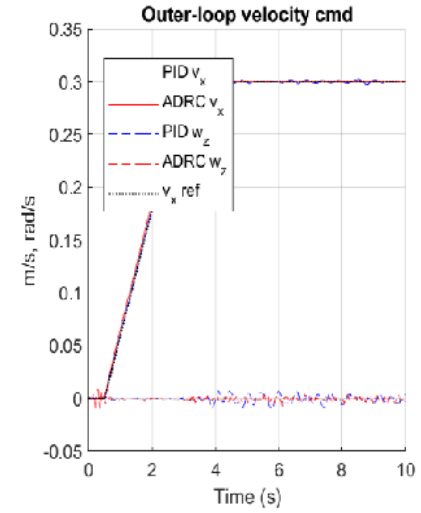
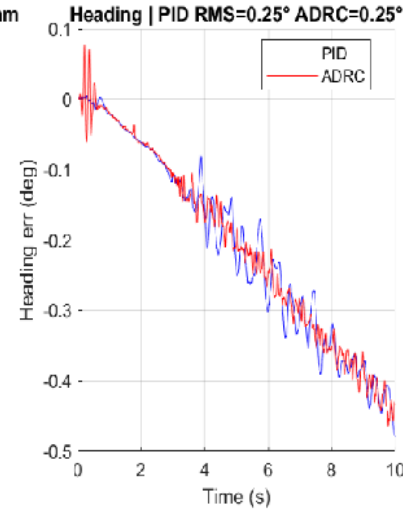
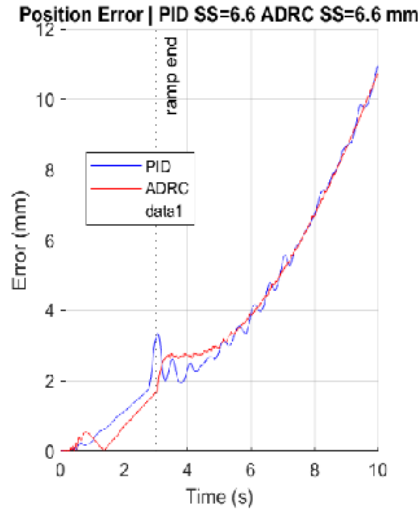
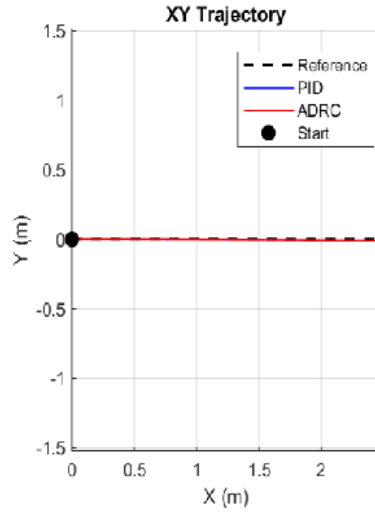
PID vs ADRC — Steady-State Tracking Error by Scenario



ADRC SS improvement (green) / degradation (red)



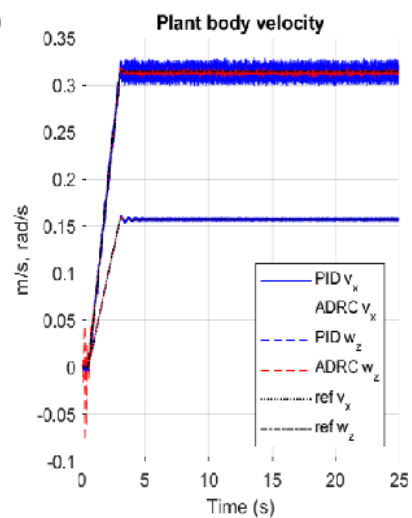
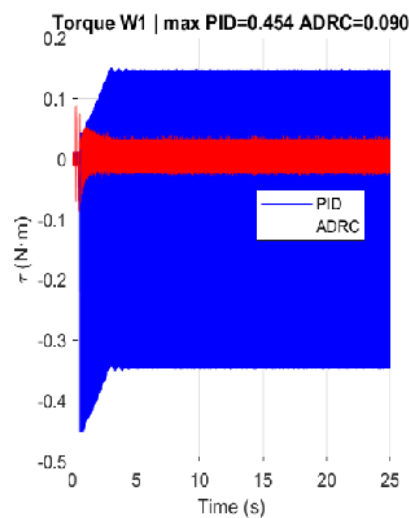
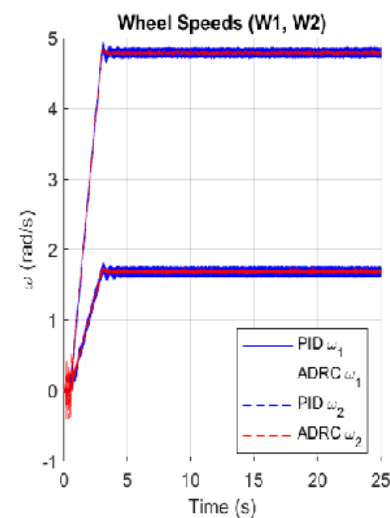
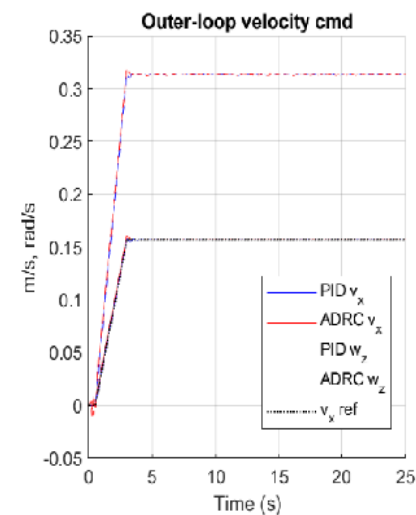
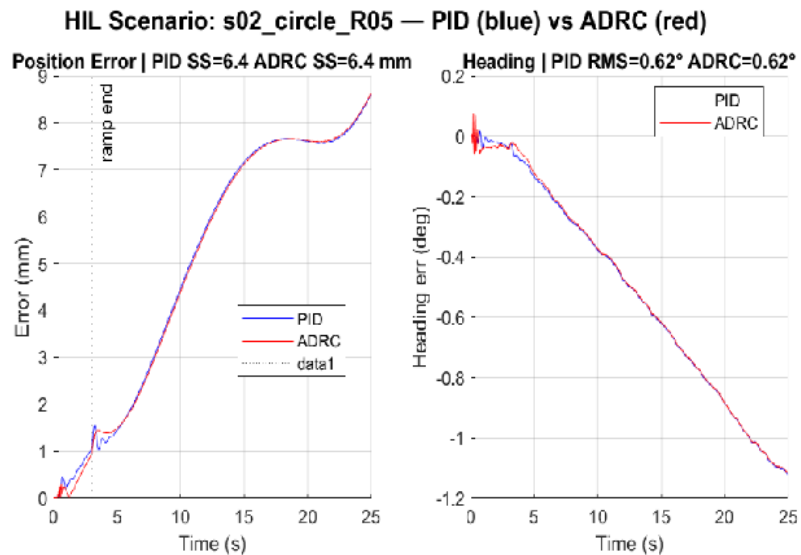
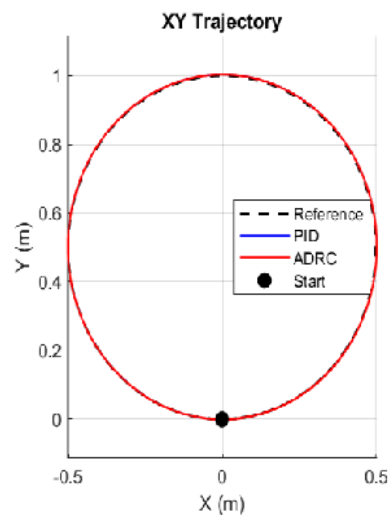
HIL Scenario: s01_line_baseline — PID (blue) vs ADRC (red)



Scenario: s01_line_baseline
Trajectory: line

Metric	PID	ADRC
RMS full (mm)	4.86	4.85
RMS SS (mm)	6.64	6.62
Peak err (mm)	10.97	10.75
t_{peak} (s)	10.00	10.00
RMS θ (°)	0.25	0.25
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.105
Slip %	0.5	0.0
Slip events	1091	554

ADRC vs PID SS: +0.3%

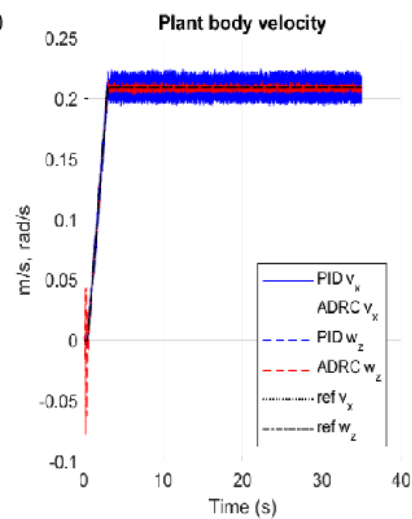
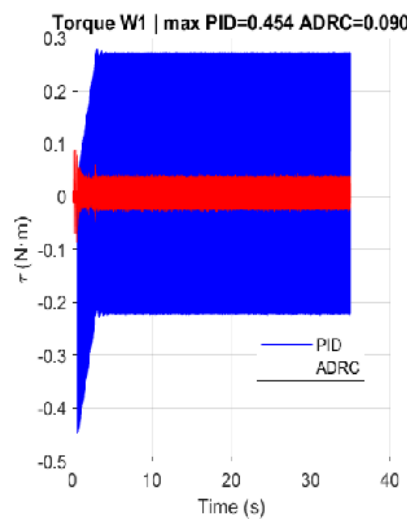
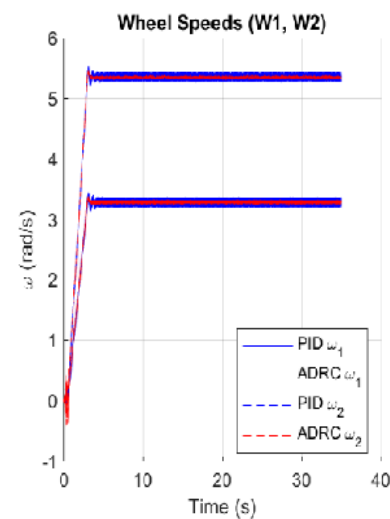
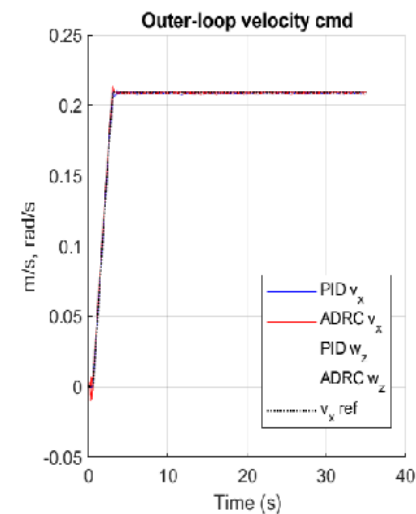
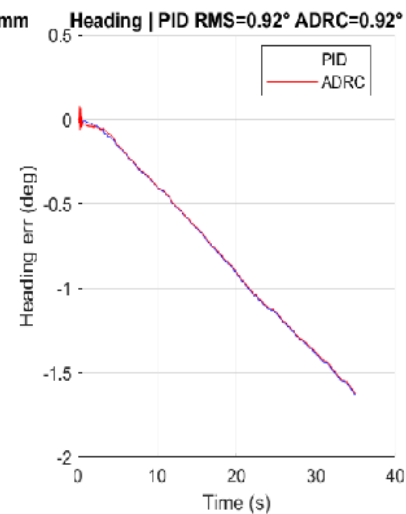
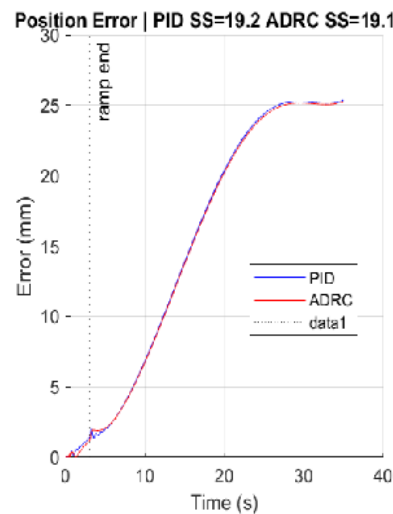
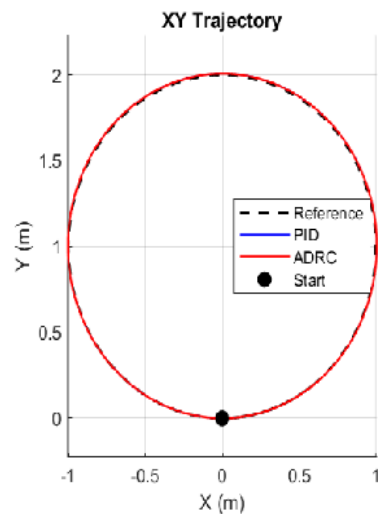


Scenario: s02_circle_R05
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	5.73	5.72
RMS SS (mm)	6.39	6.38
Peak err (mm)	8.62	9.63
t _{peak} (s)	25.00	25.00
RMS θ (°)	0.62	0.62
Settle (s)	0.00	0.00
Max τ (N·m)	0.454	0.090
Slip %	0.0	0.0
Slip events	1195	10767

ADRC vs PID SS: +0.3%

HIL Scenario: s03_circle_R10 — PID (blue) vs ADRC (red)

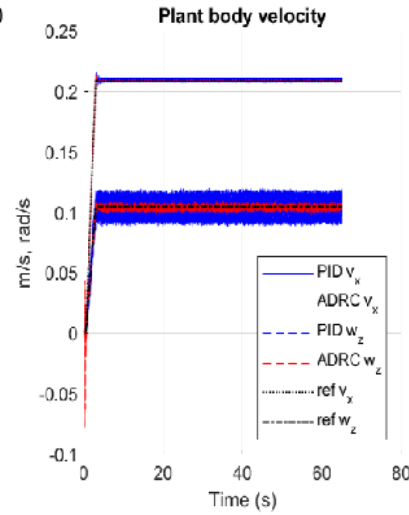
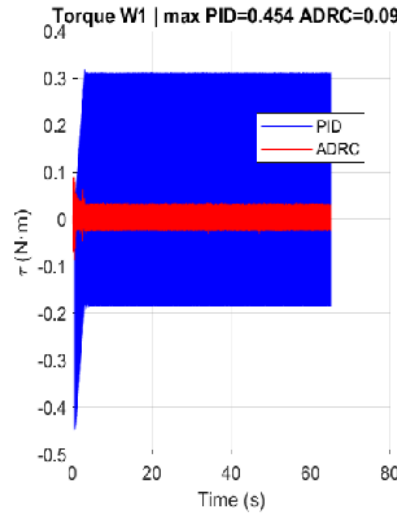
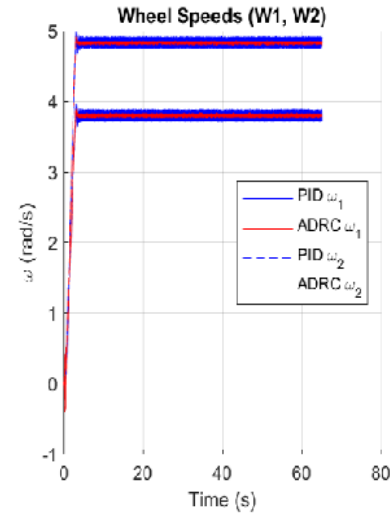
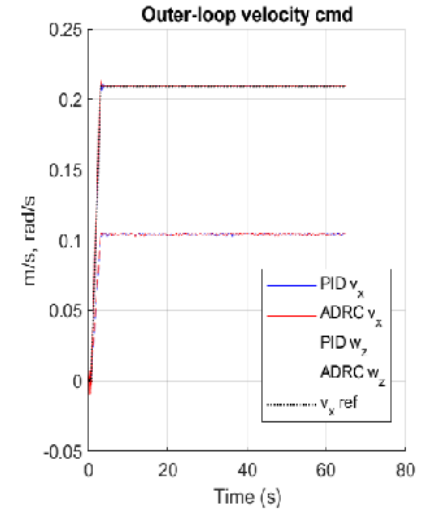
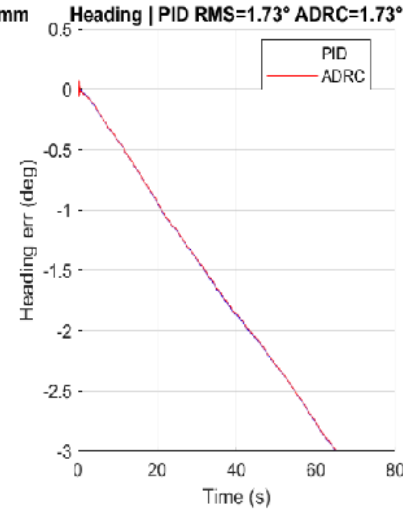
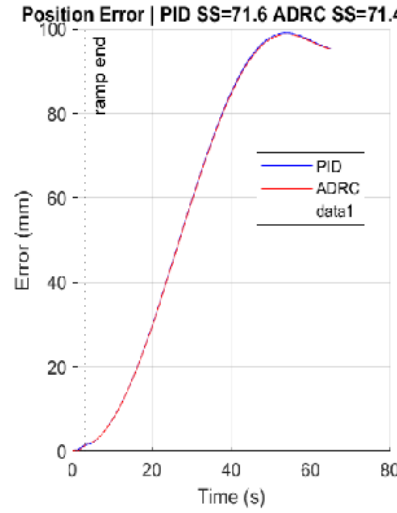
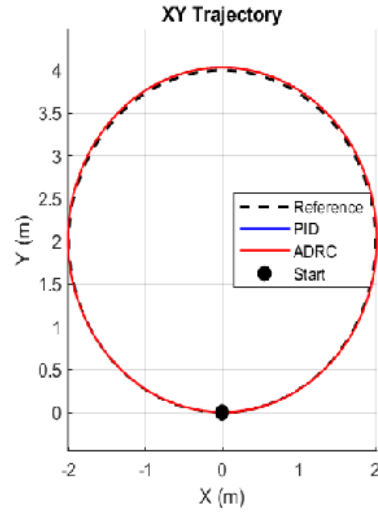


Scenario: s03_circle_R10
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	17.77	17.65
RMS SS (mm)	19.18	19.06
Peak err (mm)	25.38	25.27
t_{peak} (s)	35.00	34.98
RMS θ (°)	0.92	0.92
Settle (s)	0.00	0.00
Max τ (N·m)	0.454	0.090
Slip %	0.0	0.0
Slip events	1270	701

ADRC vs PID SS: +0.6%

HIL Scenario: s04_circle_R20 — PID (blue) vs ADRC (red)

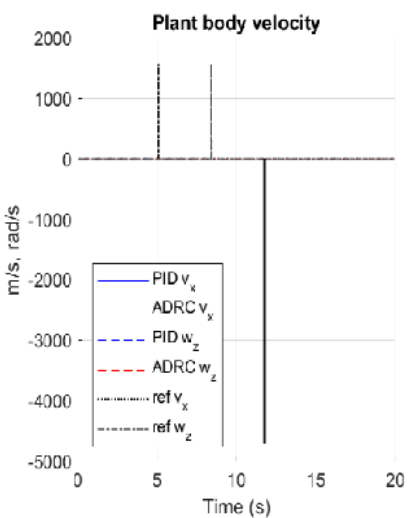
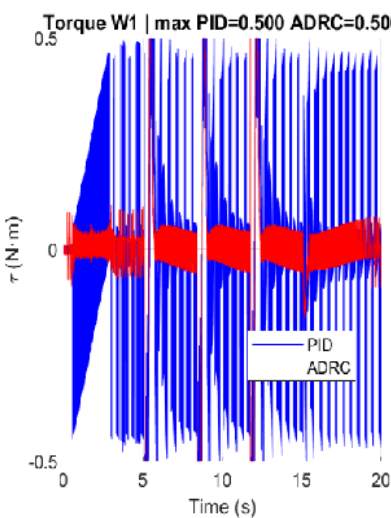
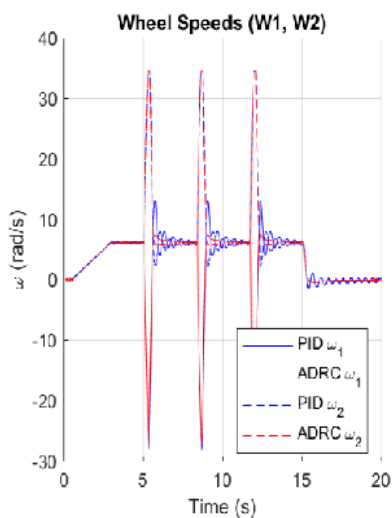
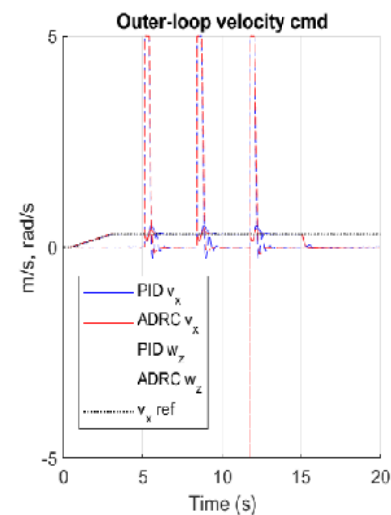
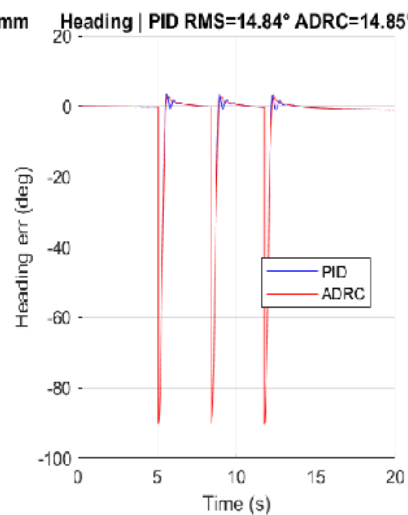
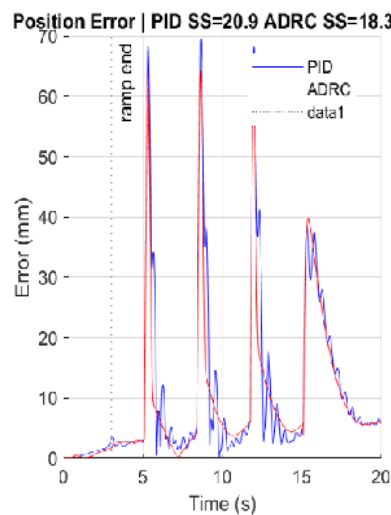
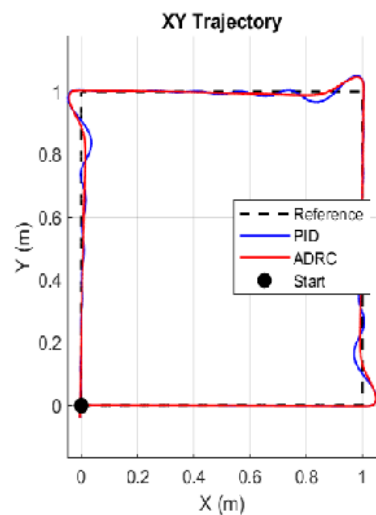


Scenario: s04_circle_R20
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	68.78	68.55
RMS SS (mm)	71.54	71.35
Peak err (mm)	99.04	99.71
t_{peak} (s)	53.87	53.90
RMS θ (°)	1.73	1.73
Settle (s)	0.00	0.00
Max τ (N·m)	0.454	0.090
Slip %	0.0	0.0
Slip events	1353	634

ADRC vs PID SS: +0.3%

HIL Scenario: s05_square — PID (blue) vs ADRC (red)

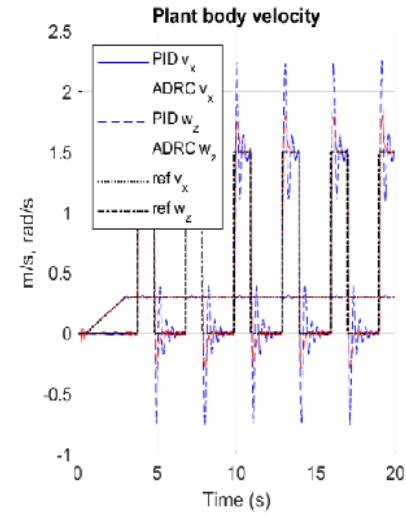
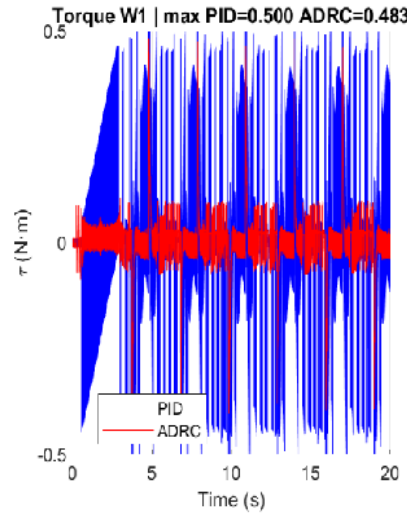
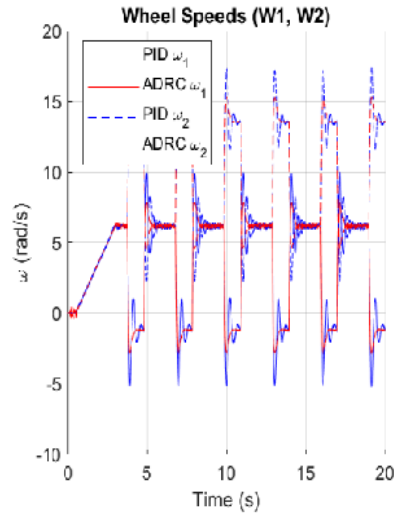
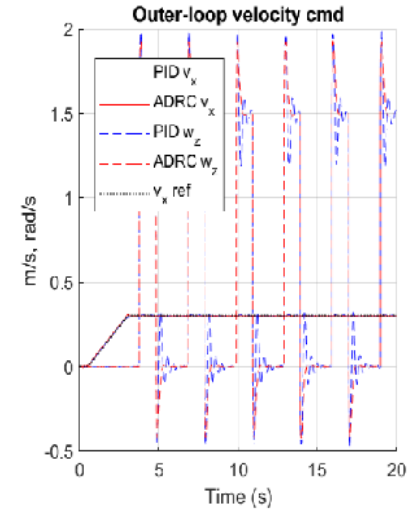
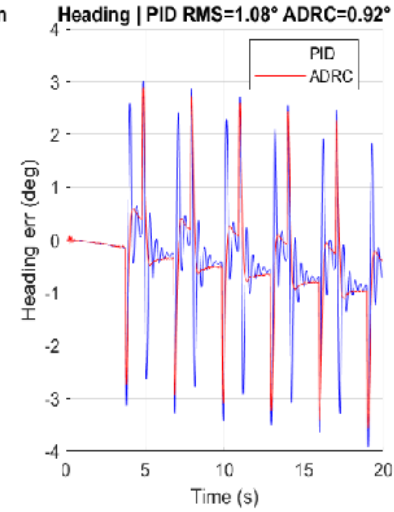
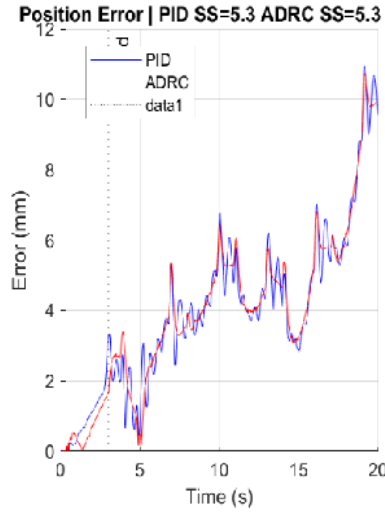
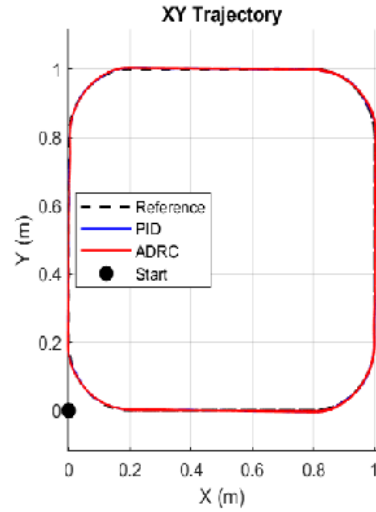


Scenario: s05_square
Trajectory: square

Metric	PID	ADRC
RMS full (mm)	18.70	15.87
RMS SS (mm)	20.88	18.28
Peak err (mm)	69.46	54.21
t_{peak} (s)	8.65	5.29
RMS θ (°)	14.84	14.85
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.500
Slip %	4.1	2.7
Slip events	6720	5060

ADRC vs PID SS: +12.4%

HIL Scenario: s06_rounded_square — PID (blue) vs ADRC (red)

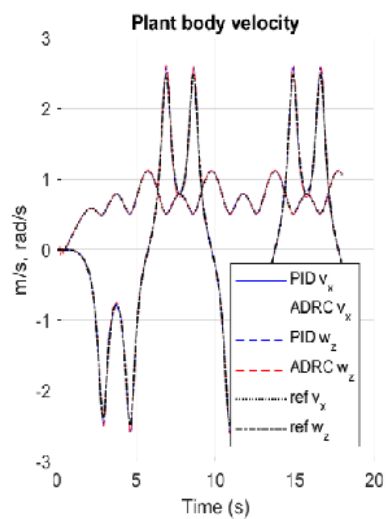
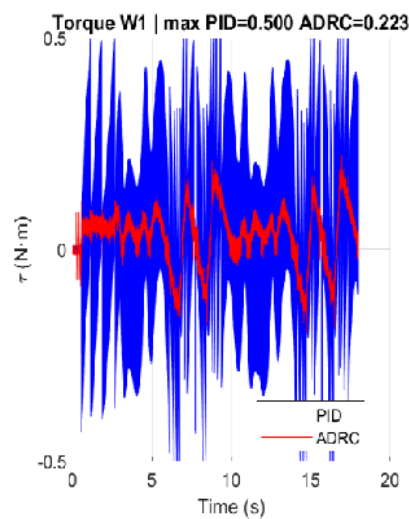
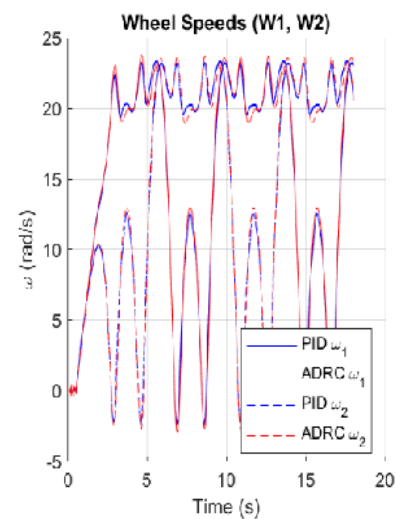
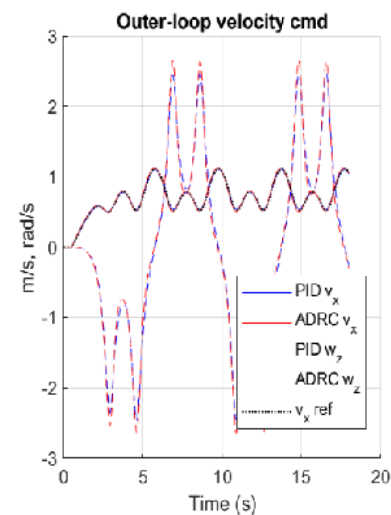
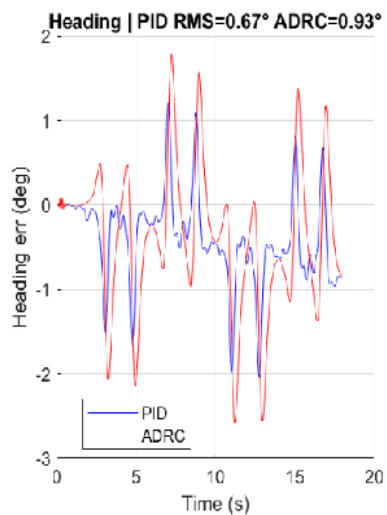
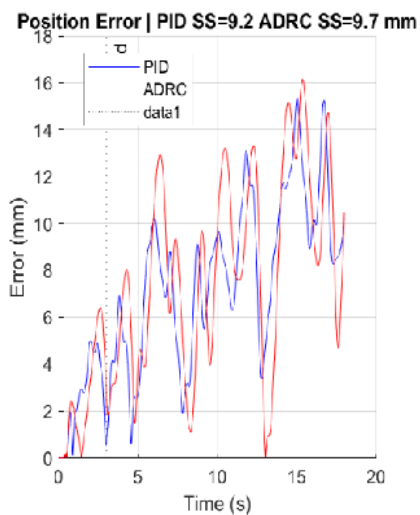
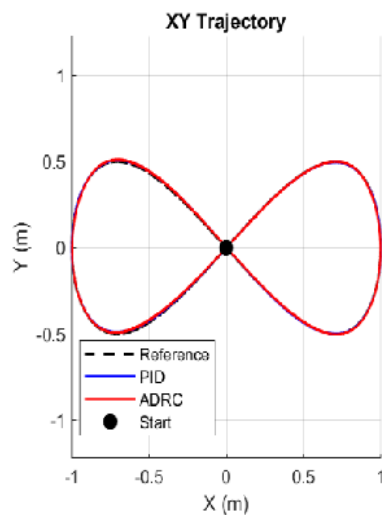


Scenario: s06_rounded_square
Trajectory: rounded_square

Metric	PID	ADRC
RMS full (mm)	4.64	4.65
RMS SS (mm)	5.28	5.30
Peak err (mm)	10.96	10.75
t_{peak} (s)	19.17	19.16
RMS θ (°)	1.08	0.92
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.483
Slip %	1.1	0.0
Slip events	3531	3747

ADRC vs PID SS: -0.4%

HIL Scenario: s07_figure8 — PID (blue) vs ADRC (red)

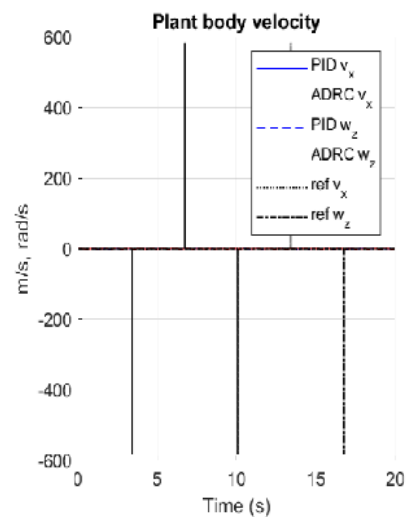
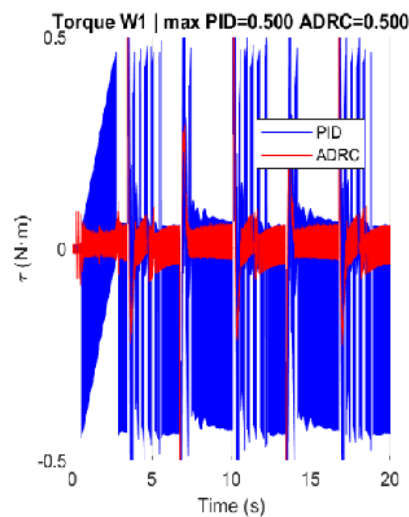
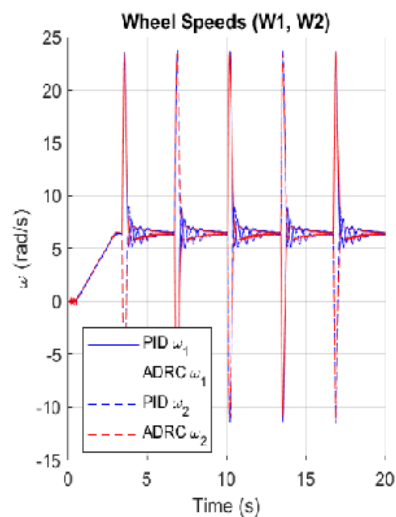
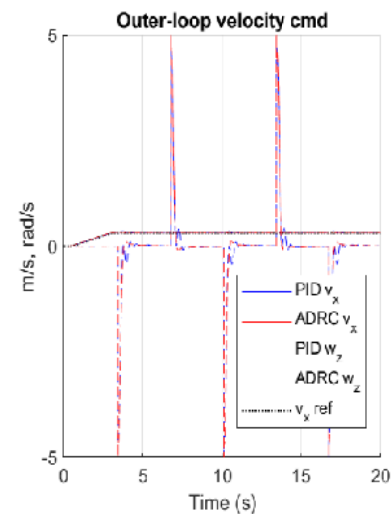
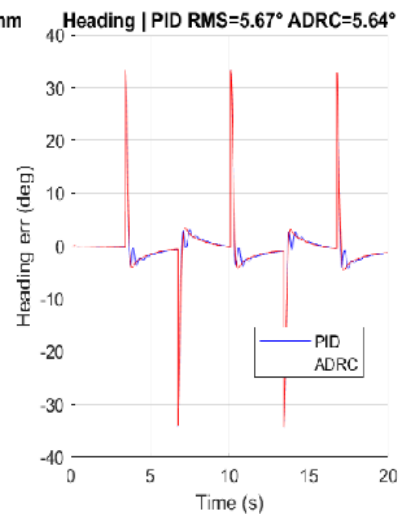
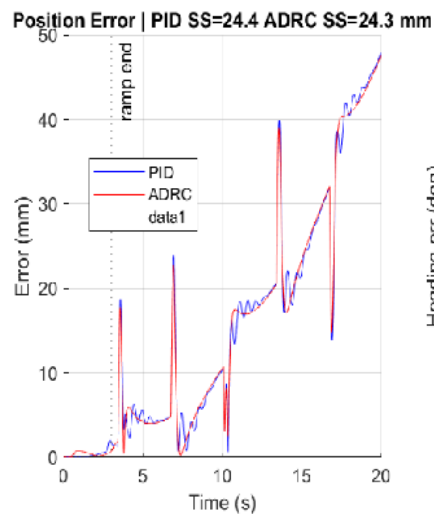
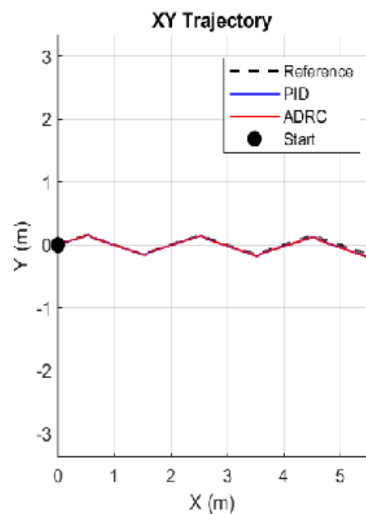


Scenario: s07_figure8
Trajectory: figure8

Metric	PID	ADRC
RMS full (mm)	8.06	8.54
RMS SS (mm)	9.23	9.73
Peak err (mm)	15.36	16.15
t _{peak} (s)	15.03	15.37
RMS θ (°)	0.67	0.93
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.223
Slip %	0.5	0.0
Slip events	1506	1943

ADRC vs PID SS: -5.5%

HIL Scenario: s08_zigzag — PID (blue) vs ADRC (red)

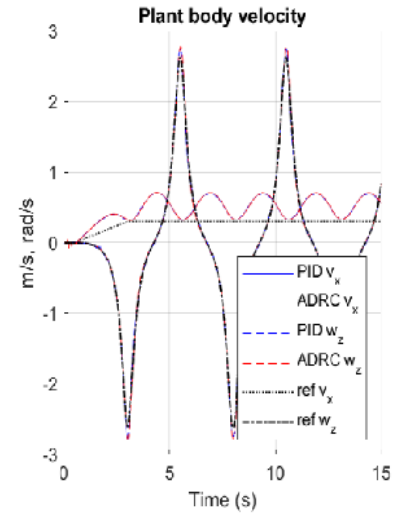
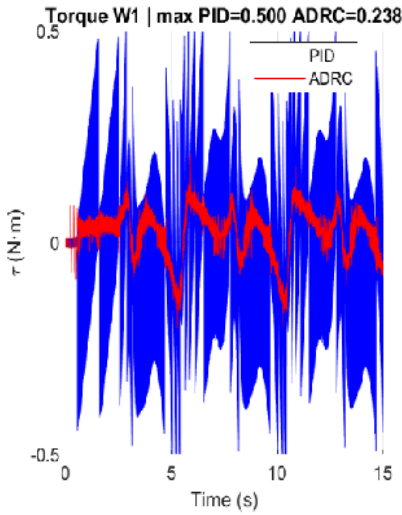
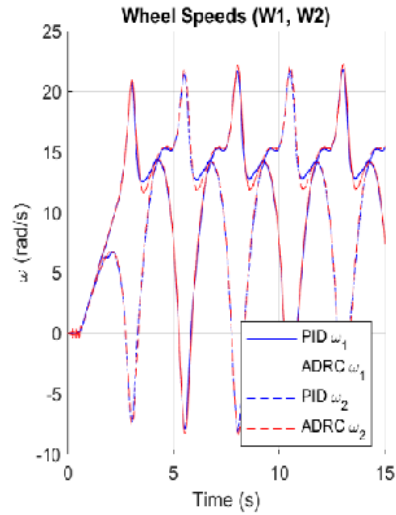
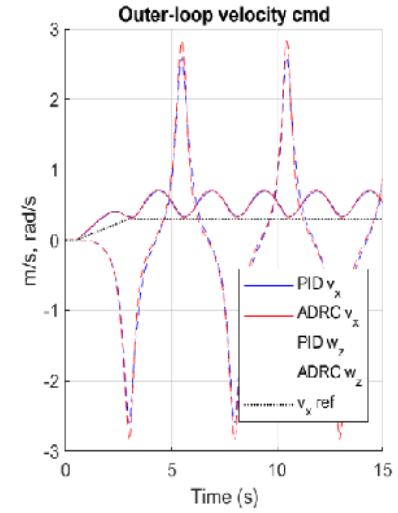
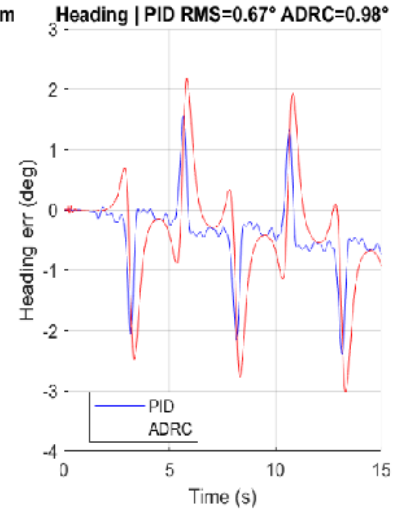
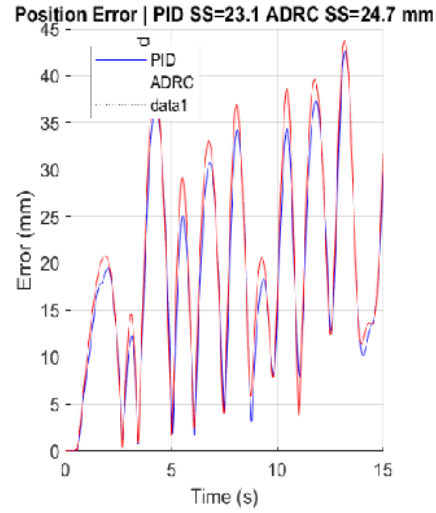
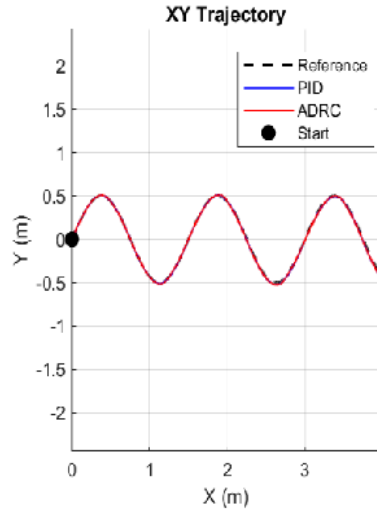


Scenario: s08_zigzag
Trajectory: zigzag

Metric	PID	ADRC
RMS full (mm)	21.20	21.12
RMS SS (mm)	24.36	24.27
Peak err (mm)	47.86	47.58
t_{peak} (s)	20.00	20.00
RMS θ (°)	5.67	5.64
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.500
Slip %	2.8	1.9
Slip events	1268	690

ADRC vs PID SS: +0.4%

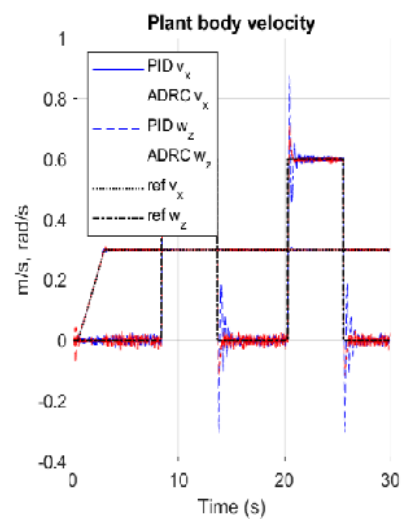
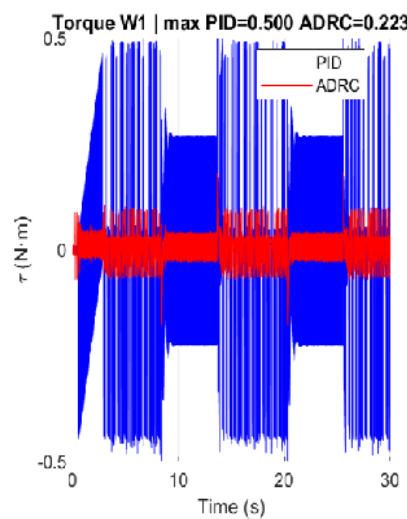
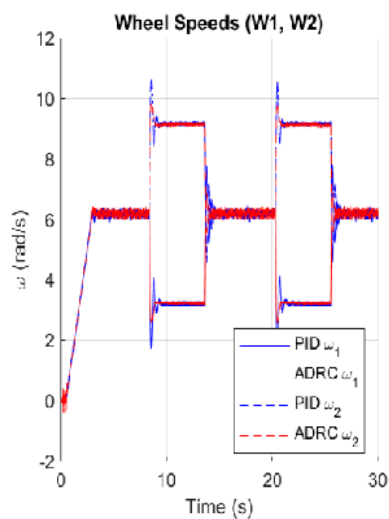
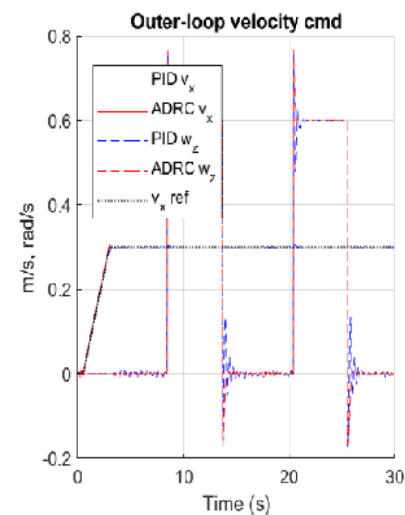
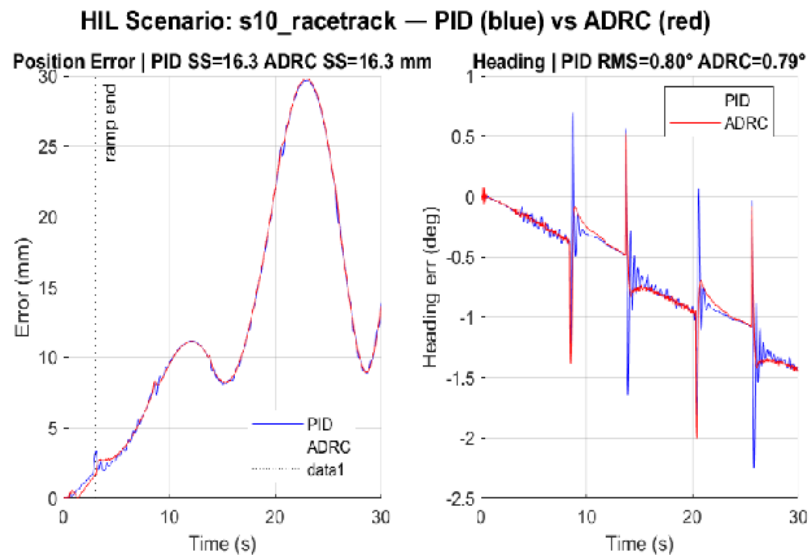
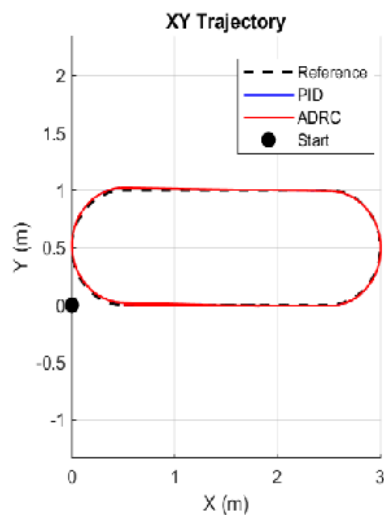
HIL Scenario: s09_sinusoidal — PID (blue) vs ADRC (red)



Scenario: s09_sinusoidal
Trajectory: sinusoidal

Metric	PID	ADRC
RMS full (mm)	21.42	22.86
RMS SS (mm)	23.08	24.71
Peak err (mm)	42.68	40.73
t_{peak} (s)	13.21	13.18
RMS θ (°)	0.67	0.98
Settle (s)	0.24	0.27
Max τ (N·m)	0.500	0.238
Slip %	0.3	0.0
Slip events	1076	1132

ADRC vs PID SS: -7.1%

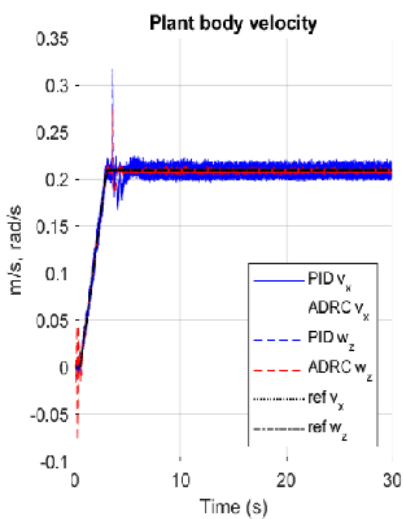
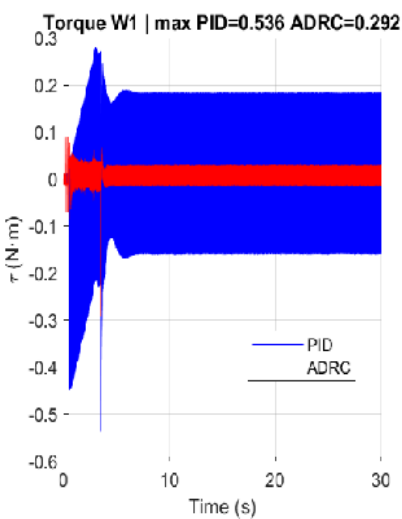
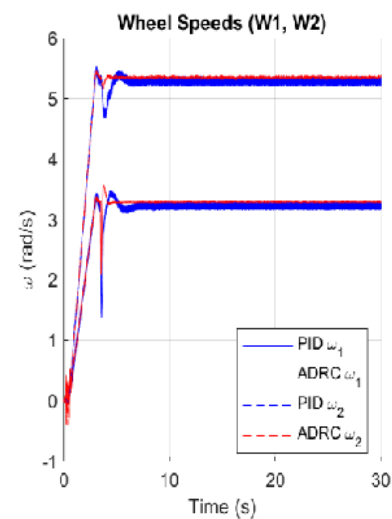
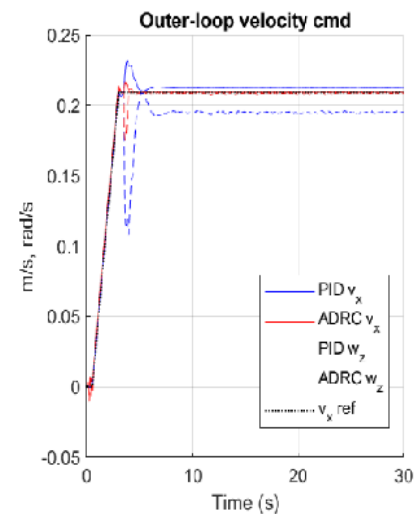
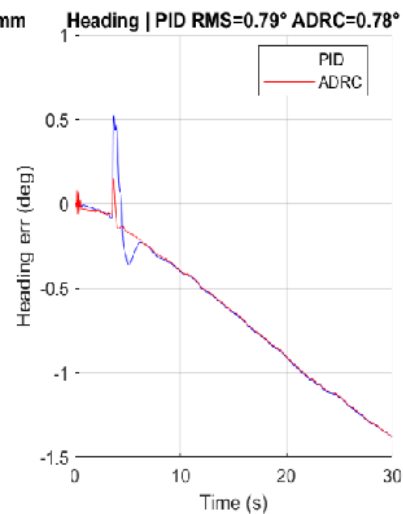
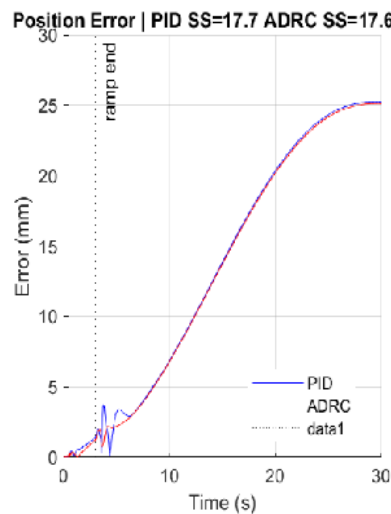
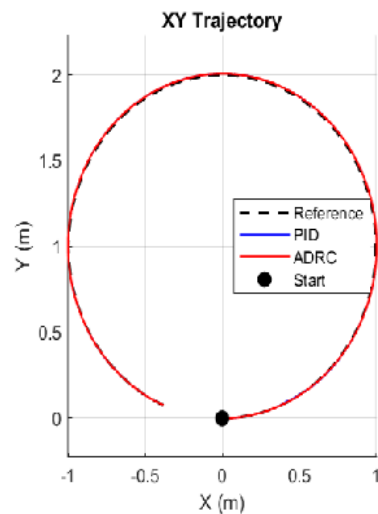


Scenario: s10_racetrack
Trajectory: racetrack

Metric	PID	ADRC
RMS full (mm)	14.85	14.90
RMS SS (mm)	16.25	16.31
Peak err (mm)	29.68	29.80
t_{peak} (s)	22.92	23.01
RMS θ (°)	0.80	0.79
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.223
Slip %	0.4	0.0
Slip events	1166	559

ADRC vs PID SS: -0.4%

HIL Scenario: s11_wheel_jam — PID (blue) vs ADRC (red)

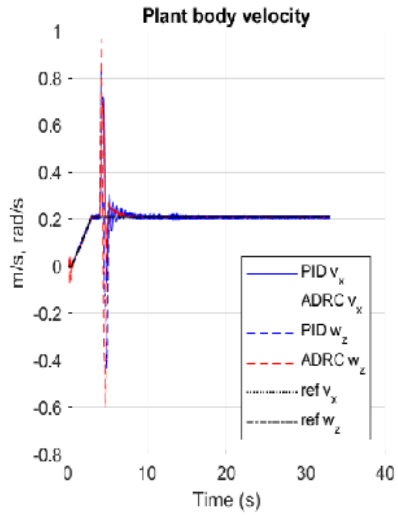
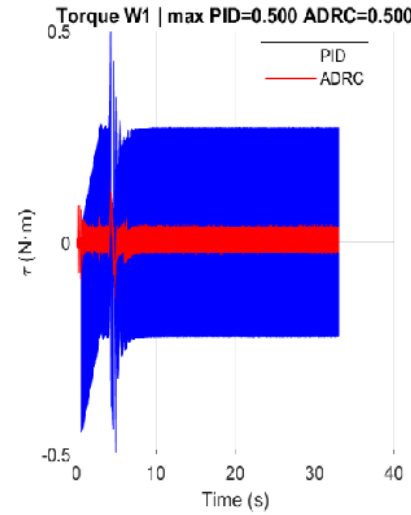
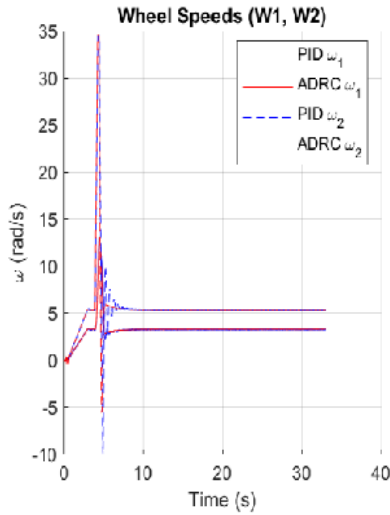
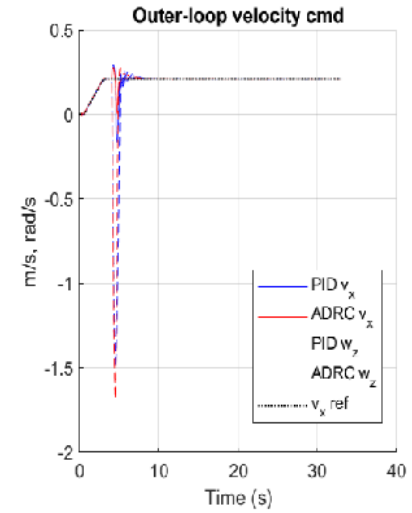
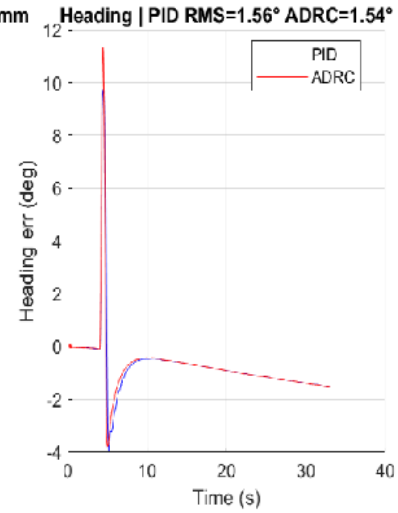
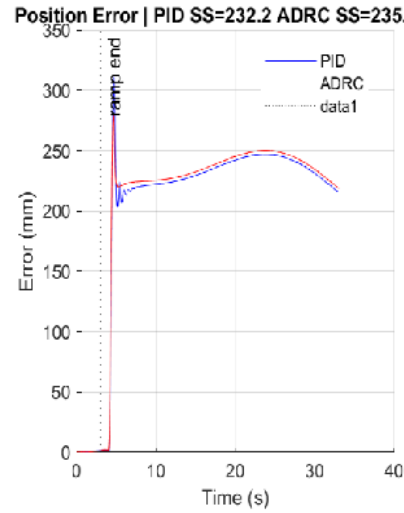
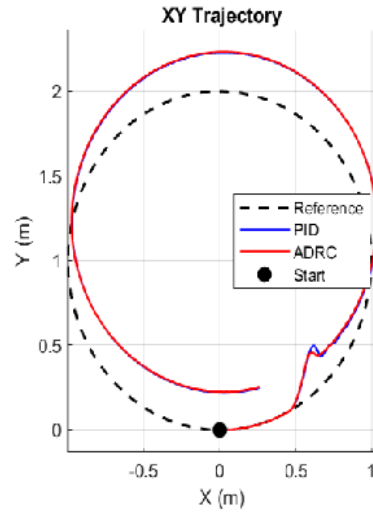


Scenario: s11_wheel_jam
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	16.7	16.05
RMS SS (mm)	17.70	17.57
Peak err (mm)	25.23	25.12
t_{peak} (s)	29.28	29.56
RMS θ (°)	0.79	0.78
Settle (s)	0.00	0.00
Max τ (N·m)	0.536	0.292
Slip %	0.0	0.0
Slip events	2044	765

ADRC vs PID SS: +0.7%

HIL Scenario: s12_encoder_dropout — PID (blue) vs ADRC (red)

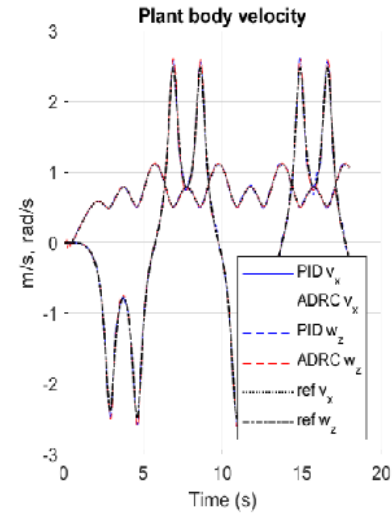
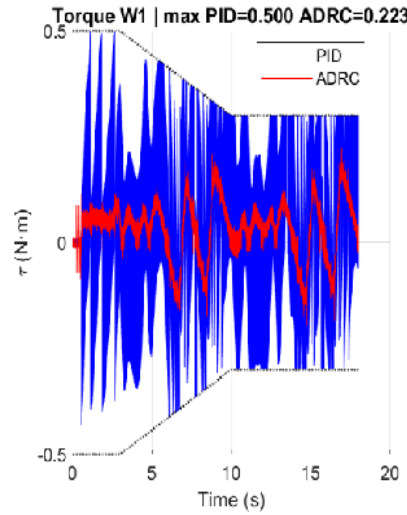
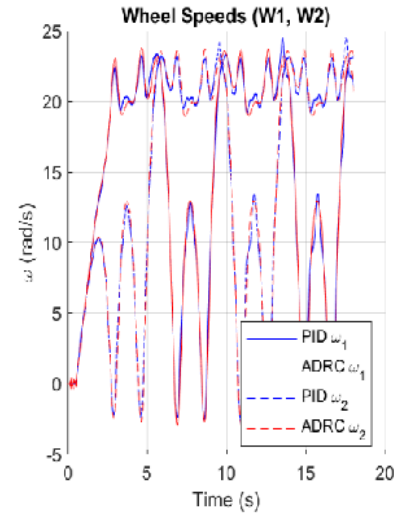
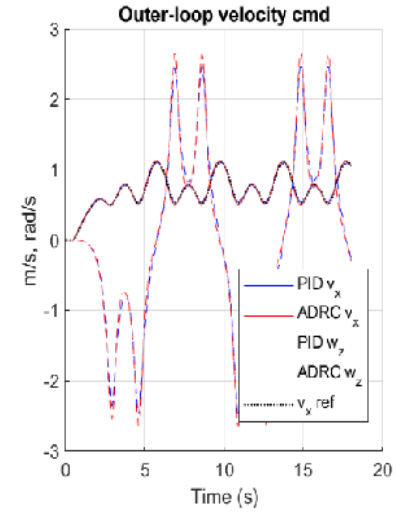
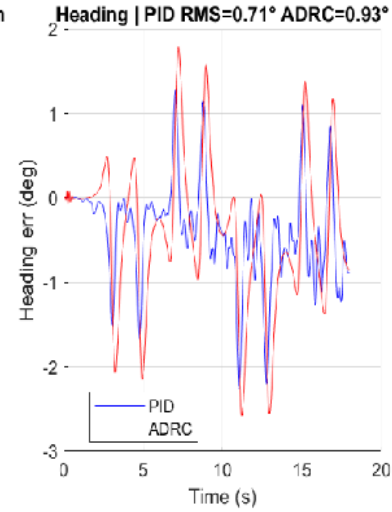
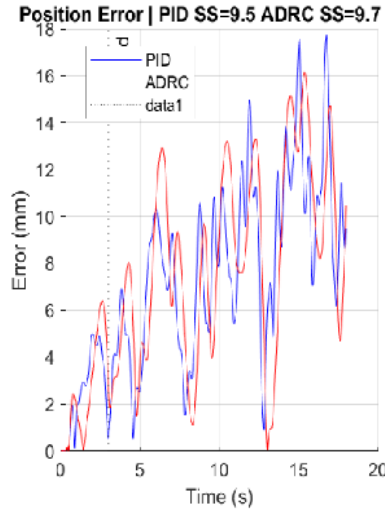
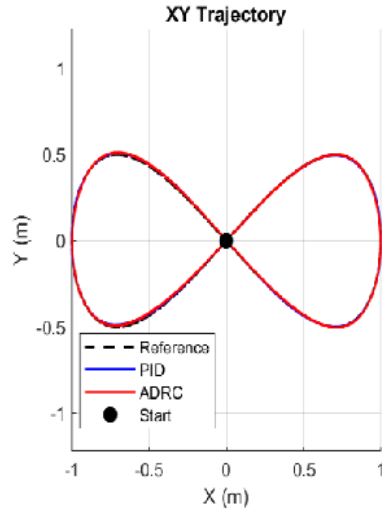


Scenario: s12_encoder_dropout
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	217.05	219.81
RMS SS (mm)	232.26	235.65
Peak err (mm)	311.17	279.07
t_{peak} (s)	4.71	4.66
RMS θ (°)	1.56	1.54
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.500
Slip %	0.7	0.4
Slip events	4623	3311

ADRC vs PID SS: -1.5%

HIL Scenario: s13_battery_fade — PID (blue) vs ADRC (red)

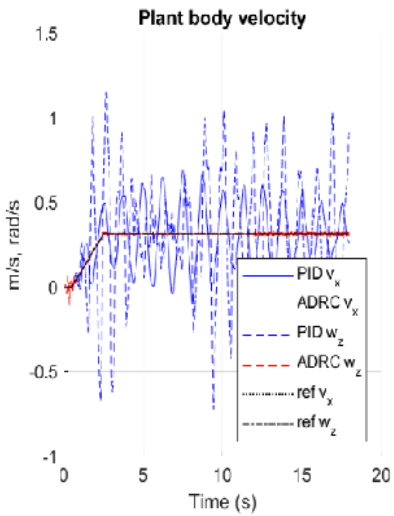
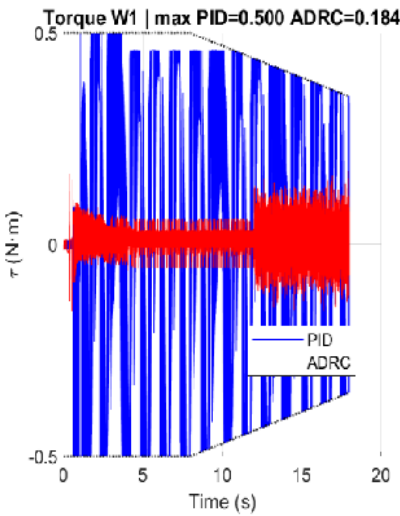
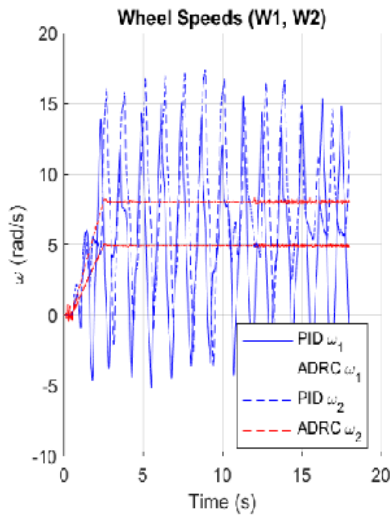
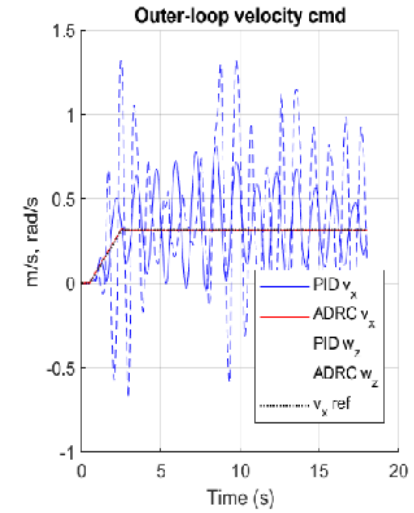
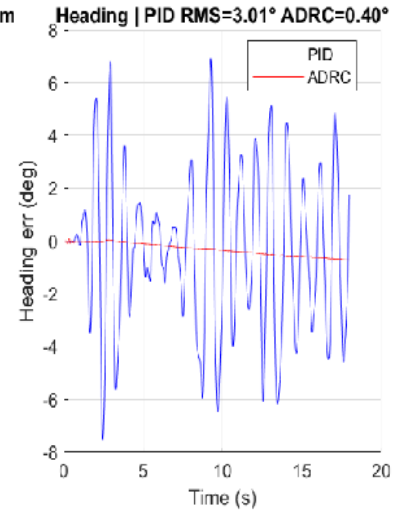
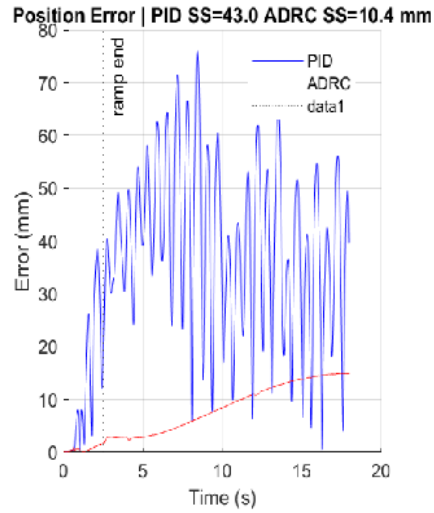
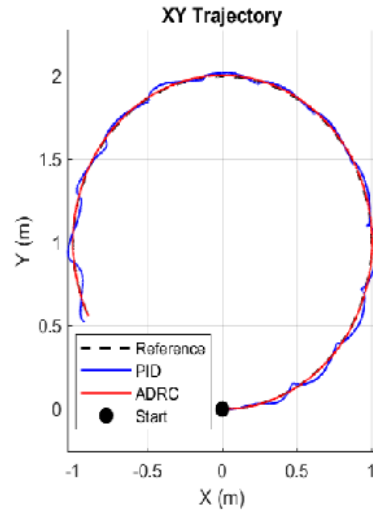


Scenario: s13_battery_fade
Trajectory: figure8

Metric	PID	ADRC
RMS full (mm)	8.30	8.54
RMS SS (mm)	9.52	9.73
Peak err (mm)	17.73	16.15
t_{peak} (s)	16.74	15.37
RMS θ (°)	0.71	0.93
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.223
Slip %	0.1	0.0
Slip events	1350	1943

ADRC vs PID SS: -2.3%

HIL Scenario: s14_industrial_nightmare — PID (blue) vs ADRC (red)

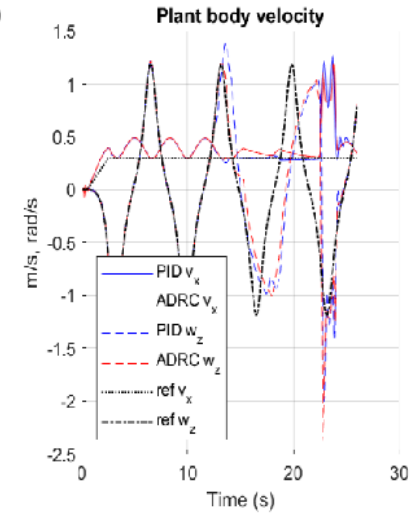
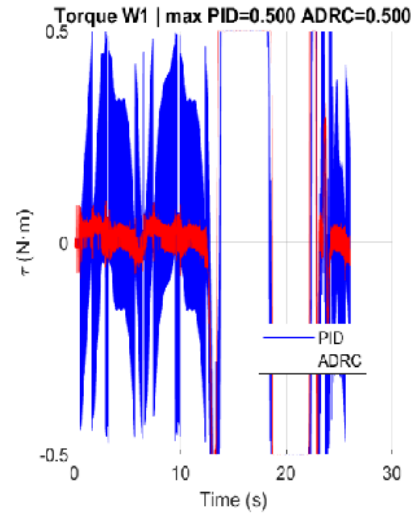
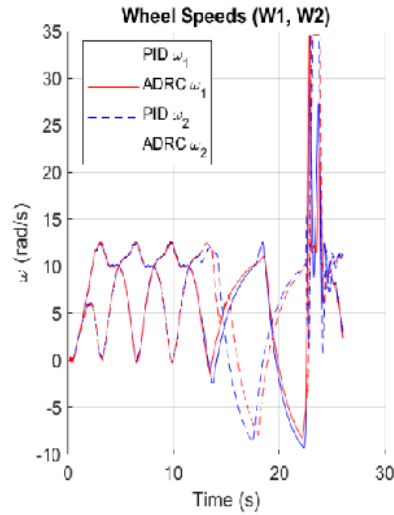
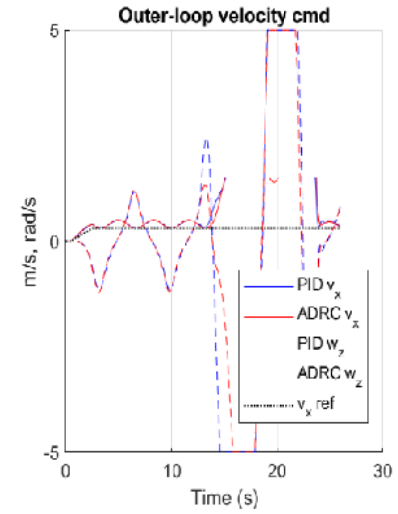
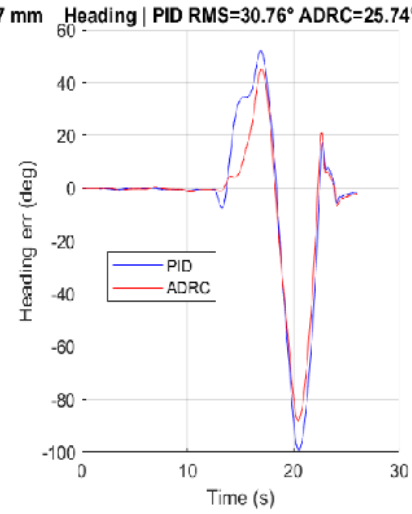
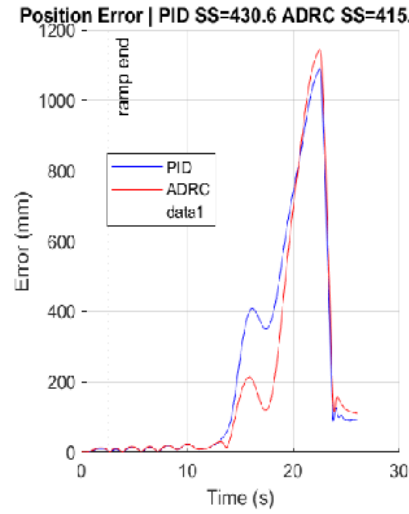
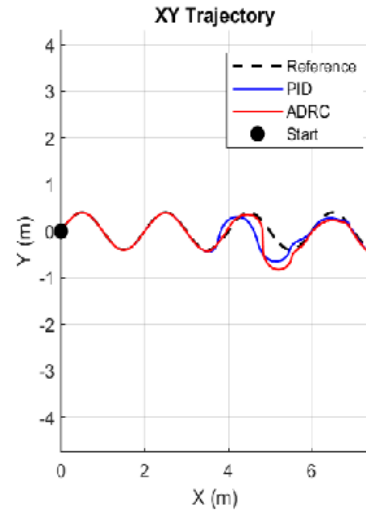


Scenario: s14_industrial_nightmare
Trajectory: circle

Metric	PID	ADRC
RMS full (mm)	39.88	9.03
RMS SS (mm)	43.05	10.37
Peak err (mm)	75.90	14.90
t_{peak} (s)	8.64	18.00
RMS θ (°)	3.01	0.40
Settle (s)	5.57	0.00
Max τ (N·m)	0.500	0.184
Slip %	13.8	0.0
Slip events	26227	1750

ADRC vs PID SS: +75.9%

HIL Scenario: s15_oily_floor — PID (blue) vs ADRC (red)



Scenario: s15_oily_floor
Trajectory: sinusoidal

Metric	PID	ADRC
RMS full (mm)	391.58	378.03
RMS SS (mm)	430.62	415.71
Peak err (mm)	1091.14	1145.54
t_{peak} (s)	22.51	22.51
RMS θ (°)	30.76	25.74
Settle (s)	0.00	0.00
Max τ (N·m)	0.500	0.500
Slip %	38.4	36.7
Slip events	38649	41029

ADRC vs PID SS: +3.5%